

MATH 8090: Review and Further Topics

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Week 16

Agenda

Review

Further Topics

Time Domain vs. Frequency Domain

Time Domain	Frequency Domain
▶ Stationarity, ACVF, and ACF ▶ Linear processes, causality, and invertibility ▶ ARMA models: estimation/forecasting ▶ ARIMA and seasonal ARIMA models	▶ Spectral density and periodogram ▶ Nonparametric spectral estimation ▶ Parametric spectral estimation ▶ Cross-spectra, coherence, and lagged regression

Two complementary views of temporal dependence

What Do We Use Time Series Analysis For?

- ▶ **Describe and understand temporal structure**

Identify trends, seasonality, dependence, cycles, and changing variability

- ▶ **Forecast the future**

Use ARMA/ARIMA and state-space models to predict future observations and quantify uncertainty

- ▶ **Estimate and test**

Estimate model parameters and assess hypotheses about temporal dependence and model structure

- ▶ **Model complex temporal behavior**

Capture volatility, frequency structure, and latent processes using GARCH, spectral, and state-space models

- ▶ **Simulate and evaluate**

Generate realistic time series from fitted models for uncertainty assessment and model evaluation

Time Series Modeling: A General Workflow

- ▶ **First, plot and explore the time series**

Look for trends, seasonality, step changes, outliers, and changes in variability

- ▶ **Transform the series to achieve approximate stationarity**

- ▶ Estimate and remove trend μ_t and seasonal component s_t
- ▶ Apply differencing when needed
- ▶ Consider nonlinear transformations (e.g., $\log$, $\sqrt{\cdot}$)

- ▶ **Model the remaining temporal dependence**

Use the ACF/PACF and other diagnostics to identify and fit an appropriate model

- ▶ **Check the fitted model**

Examine residuals to determine whether the remaining series is approximately white noise

Stationarity

$\{Y_t\}$ is **strictly stationary** if, for all $k, t_1, \dots, t_k, y_1, \dots, y_k$ and h ,

$$\mathbb{P}(Y_{t_1} \leq y_1, \dots, Y_{t_k} \leq y_k) = \mathbb{P}(Y_{t_1+h} \leq y_1, \dots, Y_{t_k+h} \leq y_k).$$

i.e., shifting the time axis does not affect the joint distribution

We consider **second-order properties**:

$\{Y_t\}$ is stationary if its **mean function** and **autocovariance function** satisfy

$$\begin{aligned}\mu_t &= \mathbb{E}[Y_t] = \mu, \\ \gamma(s, t) &= \text{Cov}(Y_s, Y_t) = \gamma(s - t).\end{aligned}$$

Note: it implies constant variance as $\gamma(t, t) = \text{Var}(Y_t) = \gamma(0)$

ACF and Sample ACF

The autocorrelation function (ACF) is

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \text{Cor}(Y_{t+h}, Y_t)$$

For observations $y_1, \dots, y_n$ of a time series, the sample mean is

$$\bar{y} = \frac{1}{n} \sum_{t=1}^n y_t.$$

The sample autocovariance function (ACVF) is

$$\hat{\gamma}(h)^1 = \frac{1}{n} \sum_{t=1}^{n-h} (y_{t+h} - \bar{y})(y_t - \bar{y}), \quad h = 0, \dots, n-1.$$

The sample autocorrelation function (ACF) is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}, \quad \hat{\rho}(-h) = \hat{\rho}(h).$$

¹At large lags, fewer observation pairs are available, so the sample ACF is less reliable.

Linear Processes

A **linear process** represents a stationary time series as a weighted combination of white-noise innovations:

$$Y_t = \mu + \sum_{j=-\infty}^{\infty} \psi_j Z_{t-j},$$

where

$$\{Z_t\} \sim \text{WN}(0, \sigma^2), \quad \sum_{j=-\infty}^{\infty} |\psi_j| < \infty.$$

Why is this important?

- ▶ The coefficients $\{\psi_j\}$ determine the **temporal dependence** of the process.
- ▶ Many key results for **ACVF/ACF, prediction, and spectral analysis** follow from the linear-process representation.
- ▶ **ARMA**(p, q) models are an important class of linear processes.

Causality and Invertibility

A linear process $\{Y_t\}$ is **causal** if there is a

$$\psi(B) = \psi_0 + \psi_1 B + \psi_2 B^2 + \dots$$

with

$$\sum_{j=0}^{\infty} |\psi_j| < \infty$$

and

$$Y_t = \psi(B)Z_t.$$

A linear process $\{Y_t\}$ is **invertible** if there is a

$$\pi(B) = \pi_0 + \pi_1 B + \pi_2 B^2 + \dots$$

with

$$\sum_{j=0}^{\infty} |\pi_j| < \infty$$

and

$$Z_t = \pi(B)Y_t.$$

Autoregressive Moving Average Models

An **ARMA**(p, q) **process** $\{Y_t\}$ is a stationary process that satisfies

$$Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p} = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q},$$

where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Also, $\phi_p, \theta_q \neq 0$ and $\phi(z)$ and $\theta(z)$ have no common factors

Properties:

- ▶ A unique **stationary** solution exists if and only if

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p = 0 \Rightarrow |z| \neq 1.$$

- ▶ This ARMA(p, q) process is **causal** if and only if

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p = 0 \Rightarrow |z| > 1.$$

- ▶ It is **invertible** if and only if

$$\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q = 0 \Rightarrow |z| > 1.$$

Linear Prediction

Given $Y_1, \dots, Y_n$, the **best linear predictor** of Y_{n+h} is

$$Y_{n+h}^n = \alpha_0 + \sum_{i=1}^n \alpha_i Y_i.$$

The prediction error is **orthogonal** to the information used for prediction:

$$\mathbb{E}[Y_{n+h} - Y_{n+h}^n] = 0, \quad \mathbb{E}[(Y_{n+h} - Y_{n+h}^n)Y_i] = 0, \quad i = 1, \dots, n.$$

One-step-ahead prediction For a zero-mean stationary time series,

$$Y_{n+1}^n = \phi_{n1}Y_n + \dots + \phi_{nn}Y_1,$$

where the prediction coefficients satisfy

$$\boxed{\Gamma_n \phi_n = \gamma_n}, \quad \gamma_n = (\gamma(1), \dots, \gamma(n))^T,$$

with Γ_n the $n \times n$ autocovariance matrix. The resulting **mean squared prediction error** is

$$\boxed{P_{n+1}^n = \gamma(0) - \gamma_n^T \Gamma_n^{-1} \gamma_n.}$$

Yule–Walker and Maximum Likelihood Estimation

Yule–Walker Estimation (AR(p))

Method of moments: match theoretical and sample autocovariances.

$$\hat{\Gamma}_p \hat{\phi} = \hat{\gamma}_p, \quad \hat{\sigma}^2 = \hat{\gamma}(0) - \hat{\phi}^T \hat{\gamma}_p.$$

Maximum Likelihood Estimation (ARMA(p, q))

For a zero-mean Gaussian ARMA process,

$$\mathbf{Y} = (Y_1, \dots, Y_n)^T \sim N(\mathbf{0}, \Gamma_n).$$

The log-likelihood is

$$\ell(\phi, \theta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\Gamma_n| - \frac{1}{2} \mathbf{Y}^T \Gamma_n^{-1} \mathbf{Y}.$$

$$(\hat{\phi}, \hat{\theta}, \hat{\sigma}^2) = \arg \max_{\phi, \theta, \sigma^2} \ell(\phi, \theta, \sigma^2)$$

ARIMA and Seasonal ARIMA Models

ARIMA(p, d, q) Difference Y_t until stationary, then model with ARMA(p, q):

$$X_t = \nabla^d Y_t = (1 - B)^d Y_t, \quad \boxed{\phi(B)(1 - B)^d Y_t = \theta(B) Z_t.}$$

Seasonal ARIMA

$$\boxed{\text{ARIMA}(p, d, q) \times (P, D, Q)_s}$$

Combines regular and seasonal differencing:

$$\nabla^d = (1 - B)^d, \quad \nabla_s^D = (1 - B^s)^D,$$

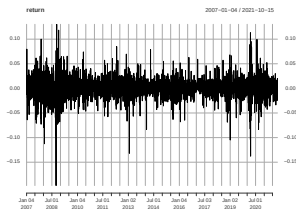
with model

$$\Phi(B^s)\phi(B)(1 - B^s)^D(1 - B)^d Y_t = \Theta(B^s)\theta(B) Z_t.$$

GARCH Models for Volatility

Financial prices are often nonstationary, so we model **log returns**

$$r_t = \log \left(\frac{y_t}{y_{t-1}} \right).$$



Returns often exhibit **volatility clustering**: large changes tend to be followed by large changes.

$$r_t = \mu_t + a_t, \quad a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2$$

Spectral Density: Frequency-Domain View

For a stationary time series with $\sum_{h=-\infty}^{\infty} |\gamma(h)| < \infty$, the **spectral density** is

$$f(\omega) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \omega h}, \quad -\frac{1}{2} \leq \omega \leq \frac{1}{2}.$$

The ACVF can be recovered from the spectrum:

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i \omega h} f(\omega) d\omega.$$

Time domain $\gamma(h) \longleftrightarrow f(\omega)$ **Frequency domain**

$$\gamma(0) = \text{Var}(Y_t) = \int_{-1/2}^{1/2} f(\omega) d\omega.$$

Periodogram and Its Properties

The **periodogram** is a frequency-domain estimator of the spectral density:

$$I(\omega_j) = |d(\omega_j)|^2 = \frac{1}{n} \left| \sum_{t=1}^n y_t e^{-2\pi i \omega_j t} \right|^2.$$

Under general conditions, for Fourier frequencies ω_j ,

$$\boxed{\frac{2I(\omega_j)}{f(\omega_j)} \xrightarrow{d} \chi_2^2}$$

and hence

$$\mathbb{E}[I(\omega_j)] \rightarrow f(\omega_j), \quad \text{Var}[I(\omega_j)] \rightarrow f^2(\omega_j).$$

Key implication: The periodogram is approximately unbiased, but **not consistent** since its variance does not vanish as $n \rightarrow \infty$.

Smoothed Periodogram

The periodogram is **highly variable**, so we smooth across nearby frequencies.

Averaged periodogram

$$\bar{f}(\omega_j) = \frac{1}{2m+1} \sum_{k=-m}^m I(\omega_{j+k}).$$

Smoothed periodogram

$$\hat{f}(\omega_j) = \sum_{k=-m}^m W_m(k) I(\omega_{j+k})$$

where $W_m(k)$ is a spectral window, typically satisfying

$$W_m(k) \geq 0, \quad W_m(k) = W_m(-k), \quad \sum_{k=-m}^m W_m(k) = 1.$$

Key idea: More smoothing $\implies$ lower variance, but potentially higher bias.

Parametric Spectral Density Estimation

Key idea: Assume a parametric time series model and use its implied spectral density. For an $AR(p)$ model:

1. Fit the model to obtain

$$\hat{\phi}_1, \dots, \hat{\phi}_p, \quad \hat{\sigma}^2.$$

2. Define the fitted AR polynomial

$$\hat{\phi}(z) = 1 - \hat{\phi}_1 z - \dots - \hat{\phi}_p z^p.$$

3. Estimate the spectrum by

$$\hat{f}(\omega) = \frac{\hat{\sigma}^2}{\left| \hat{\phi}(e^{-2\pi i \omega}) \right|^2}.$$

Parametric: model-based spectrum
smoothed periodogram

vs.

Nonparametric:

Cross-Spectrum and Coherence

For two stationary time series X_t and Y_t , the **cross-spectrum** is

$$f_{XY}(\omega) = \sum_{h=-\infty}^{\infty} \gamma_{XY}(h) e^{-2\pi i \omega h},$$

with

$$\gamma_{XY}(h) = \int_{-1/2}^{1/2} f_{XY}(\omega) e^{2\pi i \omega h} d\omega.$$

The **squared coherence** is

$$\rho_{YX}^2(\omega) = \frac{|f_{YX}(\omega)|^2}{f_X(\omega)f_Y(\omega)}, \quad 0 \leq \rho_{YX}^2(\omega) \leq 1.$$

Interpretation: Coherence measures the strength of the **linear association** between X_t and Y_t at frequency ω .

Lagged Regression in the Frequency Domain

Lagged regression

$$Y_t = \sum_{j=-\infty}^{\infty} \beta_j X_{t-j} + V_t.$$

In the frequency domain, the regression coefficients are represented by the **transfer function** $B(\omega)$:

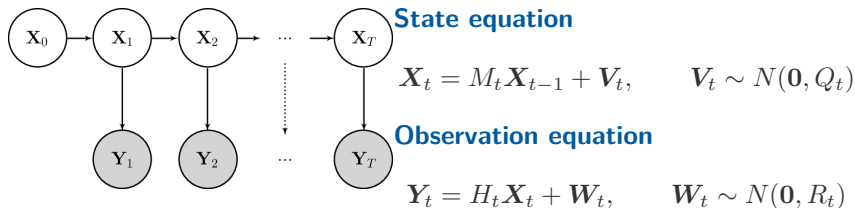
$$B(\omega) = \frac{f_{YX}(\omega)}{f_X(\omega)}$$

The minimum mean squared error is

$$\text{MSE} = \int_{-1/2}^{1/2} f_Y(\omega) [1 - \rho_{Y,X}^2(\omega)] d\omega.$$

Key interpretation: $\rho_{Y,X}^2(\omega)$ measures the proportion of Y 's variation at frequency ω explained by X .

State-Space Model



- ▶ X_t : latent **state**
- ▶ Y_t : observed data
- ▶ M_t : state **evolution**
- ▶ H_t : links state to **observations**

Key idea: latent dynamic process $\longrightarrow$ noisy observations

Forecasting, Filtering, and Smoothing

Goal: Estimate the latent state X_t using observations $Y_{1:s}$.

Available data	Problem	Interpretation
$s < t$	Forecasting	Estimate a future state
$s = t$	Filtering	Estimate the current state
$s > t$	Smoothing	Estimate a past state

Past data $\longrightarrow$ Forecast $\longrightarrow$ Filter $\longrightarrow$ Smooth

Kalman filter: forecasting + filtering

Kalman smoother: smoothing

Kalman Filter: Forecast and Update

Suppose

$$[\mathbf{X}_{t-1} \mid \mathbf{Y}_{1:t-1}] \sim N(\boldsymbol{\mu}_{t-1}^a, \Sigma_{t-1}^a).$$

1. Forecast using the state equation

$$\boldsymbol{\mu}_t^f = M_t \boldsymbol{\mu}_{t-1}^a, \quad \Sigma_t^f = M_t \Sigma_{t-1}^a M_t^T + Q_t.$$

2. Update using the new observation

$$\boxed{\boldsymbol{\mu}_t^a = \boldsymbol{\mu}_t^f + K_t \underbrace{(\mathbf{Y}_t - H_t \boldsymbol{\mu}_t^f)}_{\text{innovation}}}$$

$$K_t = \Sigma_t^f H_t^T \left(H_t \Sigma_t^f H_t^T + R_t \right)^{-1}, \quad \Sigma_t^a = (I - K_t H_t) \Sigma_t^f.$$

Kalman gain K_t : balances model forecast and new information.

Multivariate Time Series

A multivariate time series consists of several variables observed together over time:

$$\mathbf{Y}_t = (Y_{1t}, \dots, Y_{pt})^T \in \mathbb{R}^p.$$

Unlike a univariate series, we model both **temporal dependence** and **dependence across variables**.

For a stationary multivariate process, define the **cross-covariance matrix**

$$\Gamma(h) = \text{Cov}(\mathbf{Y}_{t+h}, \mathbf{Y}_t),$$

whose (i, j) entry is

$$\gamma_{ij}(h) = \text{Cov}(Y_{i,t+h}, Y_{j,t}).$$

$$\textbf{Univariate: } \gamma(h) \quad \longrightarrow \quad \textbf{Multivariate: } \Gamma(h)$$

Vector Autoregressive (VAR) Models

A $\text{VAR}(k)$ model extends the univariate AR model to a vector time series:

$$\mathbf{Y}_t = \boldsymbol{\mu} + A_1 \mathbf{Y}_{t-1} + \cdots + A_k \mathbf{Y}_{t-k} + \mathbf{W}_t.$$

where

$$\mathbf{Y}_t \in \mathbb{R}^p, \quad A_j \in \mathbb{R}^{p \times p}, \quad \text{Cov}(\mathbf{W}_t) = \Sigma_W.$$

- ▶ Each variable may depend on **its own past** and the **past of other variables**.
- ▶ The matrices $A_1, \dots, A_k$ describe these dynamic relationships.
- ▶ Σ_W allows the innovations of different series to be contemporaneously correlated.

AR model $\longrightarrow$ **VAR model: jointly model multiple time series**

VARMA Models: An Identifiability Issue

In multivariate models, different AR and MA representations can describe the **same process**.

VAR(1):

$$\begin{bmatrix} Y_{1t} \\ Y_{2t} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} Y_{1,t-1} \\ Y_{2,t-1} \end{bmatrix} + \begin{bmatrix} W_{1t} \\ W_{2t} \end{bmatrix}.$$

Equivalent VMA(1):

$$\begin{bmatrix} Y_{1t} \\ Y_{2t} \end{bmatrix} = \begin{bmatrix} W_{1t} \\ W_{2t} \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} W_{1,t-1} \\ W_{2,t-1} \end{bmatrix}.$$

Indeed,

$$Y_{1t} = W_{1t}, \quad Y_{2t} = 2W_{1,t-1} + W_{2t}.$$

Key issue: VARMA representations require additional **identifiability restrictions** to obtain a unique representation.

Long-Memory Processes

A stationary process $\{Y_t\}$ exhibits **long memory** if its dependence decays slowly with lag. A common characterization is

$$\gamma(h) = \text{Cov}(Y_t, Y_{t+h}) \sim c h^{2d-1}, \quad h \rightarrow \infty,$$

where $0 < d < 1/2$ is the **long-memory parameter**.

- ▶ The ACF decays at a **power-law rate**, much more slowly than the exponential decay typical of short-memory ARMA processes.
- ▶ Dependence can remain substantial even at **very long lags**.
- ▶ A realization may show persistent periods of values above or below the mean, producing **long-range persistence**.

Short memory: rapid ACF decay $\longleftrightarrow$ **Long memory:** slow ACF decay

ARFIMA Models

Recall the ARIMA model:

$$\phi(B)(1 - B)^d Y_t = \theta(B)Z_t,$$

where d is an **integer** differencing parameter.

Allowing d to be **real-valued** gives the **ARFIMA**(p, d, q) model:

$$\boxed{\phi(B)(1 - B)^d Y_t = \theta(B)Z_t.}$$

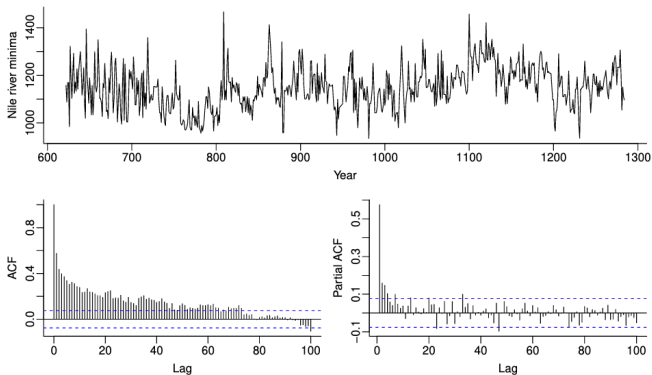
The fractional difference operator is defined through

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k.$$

- ▶ $0 < d < 1/2$: stationary **long-memory** behavior
- ▶ $d = 0$: reduces to an ARMA(p, q) model
- ▶ $d \geq 1/2$: the process is no longer stationary

Example: Nile River Minima

Annual minimum water levels of the Nile River from 622–1281, measured at the Roda gauge near Cairo (Tousson, 1925).



Feature of interest: persistent periods of relatively **high or low values** suggest dependence operating over long time scales.

Bootstrap Methods for Time Series

For independent data, bootstrap samples are generated by resampling individual observations. For time series, this can **destroy temporal dependence**.

Key idea: Resample the data in a way that **preserves dependence**.

- ▶ **Moving block bootstrap:** resample blocks of consecutive observations
- ▶ **Stationary bootstrap:** resample blocks with random lengths to produce a stationary bootstrap series

Selected references

- ▶ Künsch (1989), “The Jackknife and the Bootstrap for General Stationary Observations”
- ▶ Politis and Romano (1994), “The Stationary Bootstrap”
- ▶ Davison and Hinkley (1997), *Bootstrap Methods and Their Application*

Bayesian Time Series Analysis

Bayesian methods provide a probabilistic framework for **parameter estimation, latent-state inference, and forecasting**.

Key idea: Combine prior information with the time-series likelihood:

$$p(\boldsymbol{\theta} \mid \mathbf{Y}) \propto p(\mathbf{Y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})$$

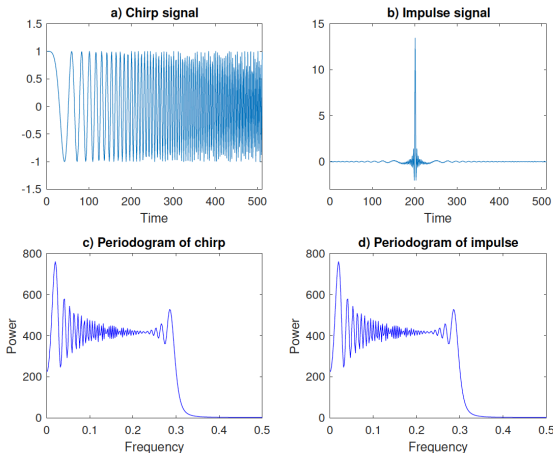
where $\boldsymbol{\theta}$ may include model parameters and latent states.

- ▶ **Uncertainty quantification:** inference is based on the full posterior distribution
- ▶ **Forecasting:** posterior predictive distributions incorporate parameter uncertainty
- ▶ **Flexible models:** natural framework for hierarchical and state-space models
- ▶ **Computation:** often relies on MCMC, particle methods, or other Bayesian algorithms

Prior + **Time-Series Likelihood** $\longrightarrow$ **Posterior** $\longrightarrow$ **Prediction**

Time-Frequency Analysis

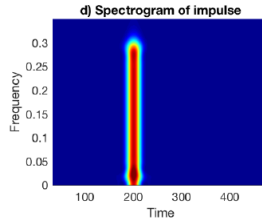
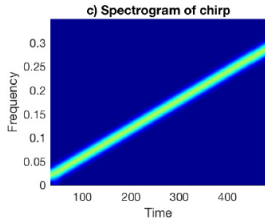
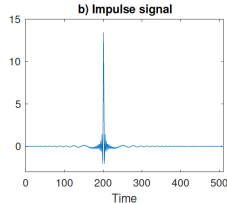
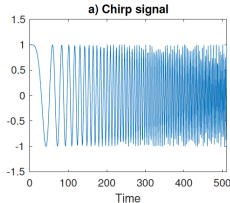
Classical spectral analysis describes the **overall frequency content** of a stationary time series $\Rightarrow$ what if the frequency **changes over time**?



Key idea: Time-frequency analysis describes **which frequencies are present and when**.

Time-Frequency Analysis: Spectrogram

A **spectrogram** estimates spectral content **locally in time** using moving time windows.



Key trade-off: Shorter window $\Rightarrow$ better **time resolution**;
window $\Rightarrow$ better **frequency resolution**.

longer

Further Topics: Non-Gaussian Time Series

Many time series are not well represented by Gaussian models, including **counts, binary outcomes, proportions, and heavy-tailed data**.

Extensions of classical time series methods include:

- ▶ Regression models for dependent non-Gaussian outcomes
- ▶ Discrete-valued and count time series
- ▶ Dynamic generalized linear models
- ▶ Bayesian approaches to non-Gaussian state-space models

Selected references

- ▶ Kedem and Fokianos (2002), *Regression Models for Time Series Analysis*
- ▶ Davis *et al.* (eds., 2016), *Handbook of Discrete-Valued Time Series*
- ▶ Gamerman *et al.* (2016), Bayesian dynamic generalized linear models
- ▶ Davis *et al.* (2021), “Count Time Series: A Methodological Review”